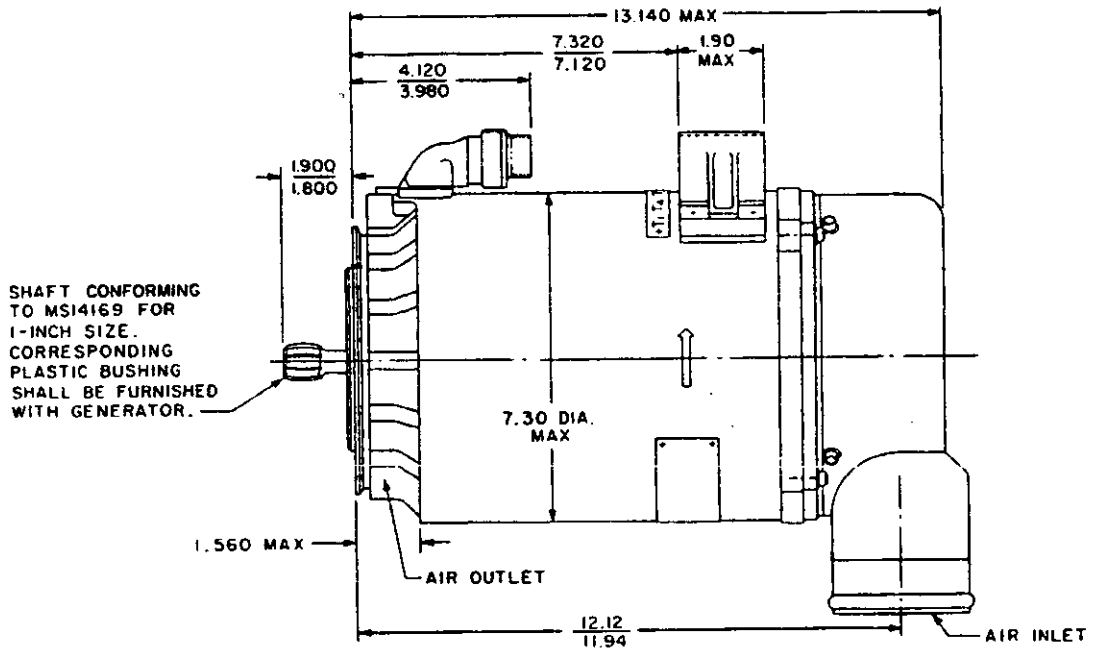
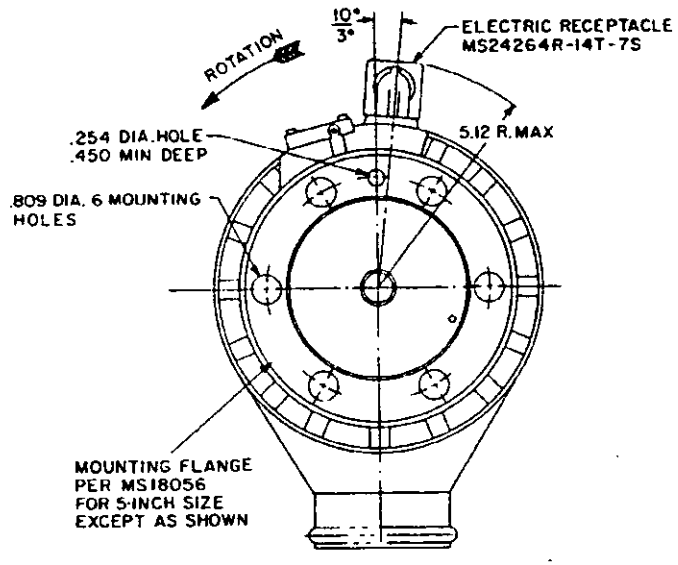
MILITARY SPECIFICATION SHEET

GENERATING SYSTEMS, AIRCRAFT, 400 HERTZ, 3-PHASE 115/200 VOLT, 50 KVA AND 90 KVA

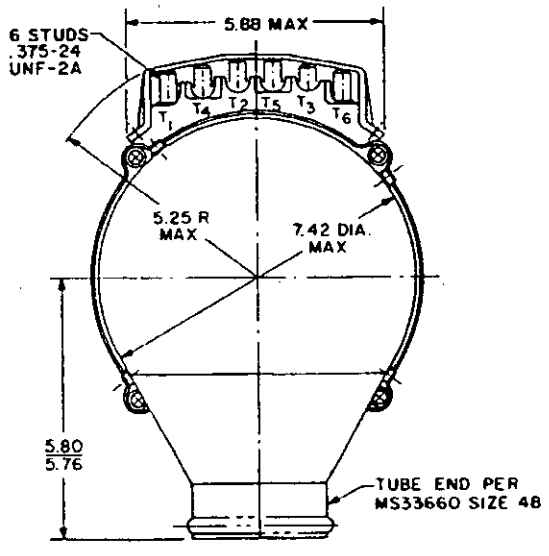
This specification is approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of specification MIL-G-21480.

This specification sheet covers two 400 hertz, 3-phase, 4-wire, 115/200 volt, grounded-neutral electric power generating systems for use on aircraft, 50 KVA and 90 KVA. The same control unit and current transformer are used with both systems (one control unit and two current transformers for each system).





MOUNTING END



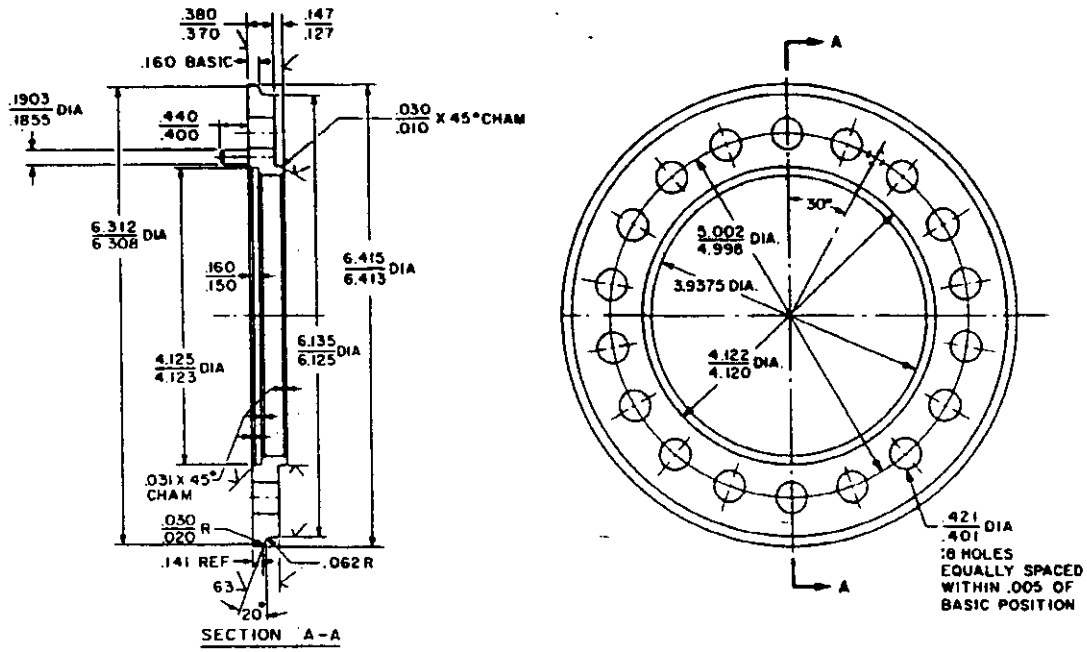
FREE END

50 KVA GENERATOR
P/N M21480/19-3

SHEET 1

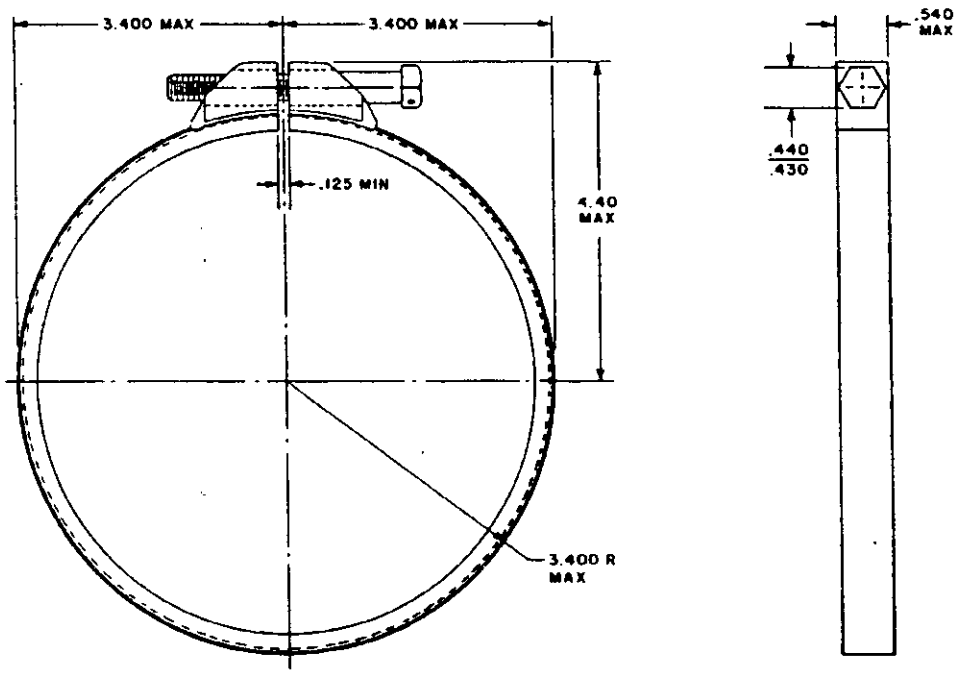
FSC 6115

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.



MOUNTING PLATE FOR 50 KVA GENERATOR

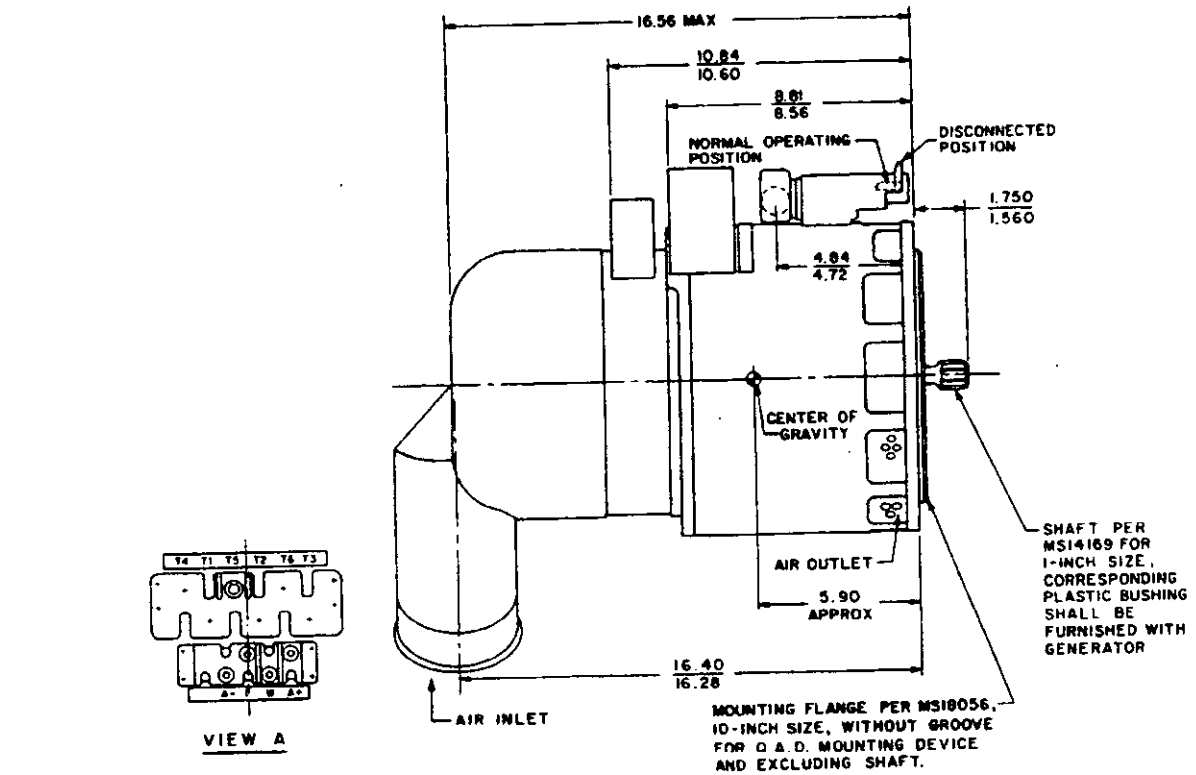
P/N M21480/19-5

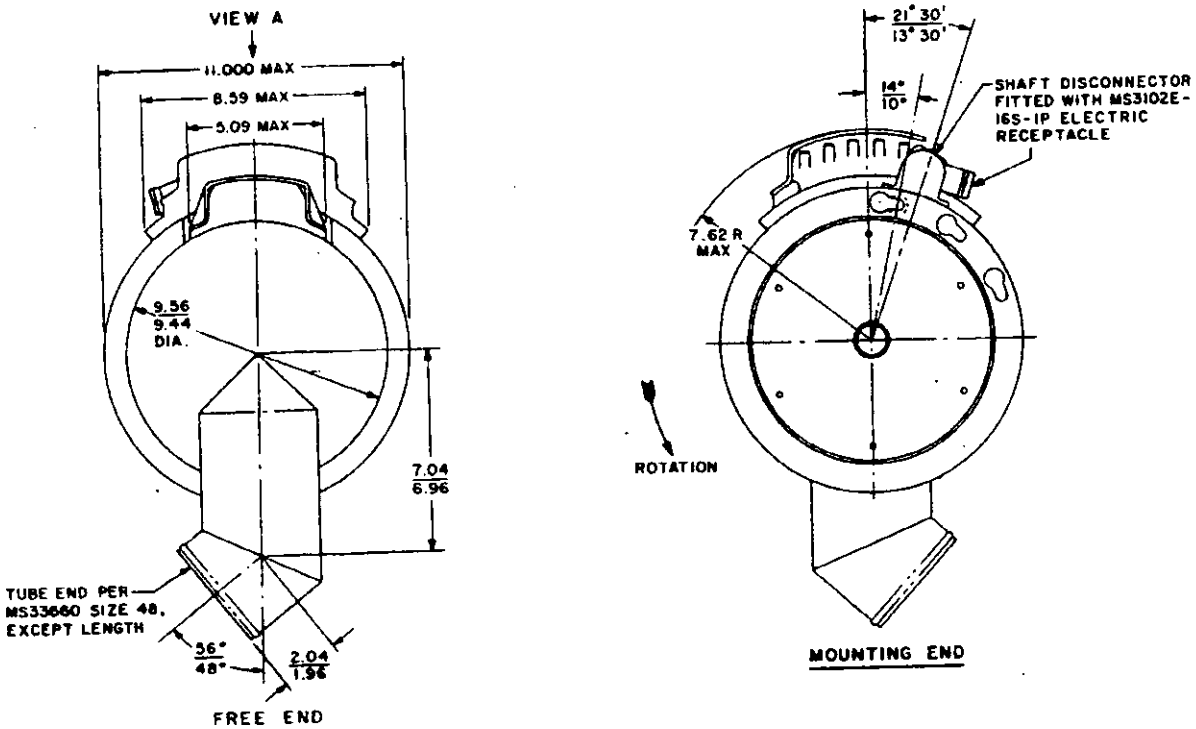


MOUNTING CLAMP FOR 50 KVA GENERATOR

P/N M21480/19-7

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

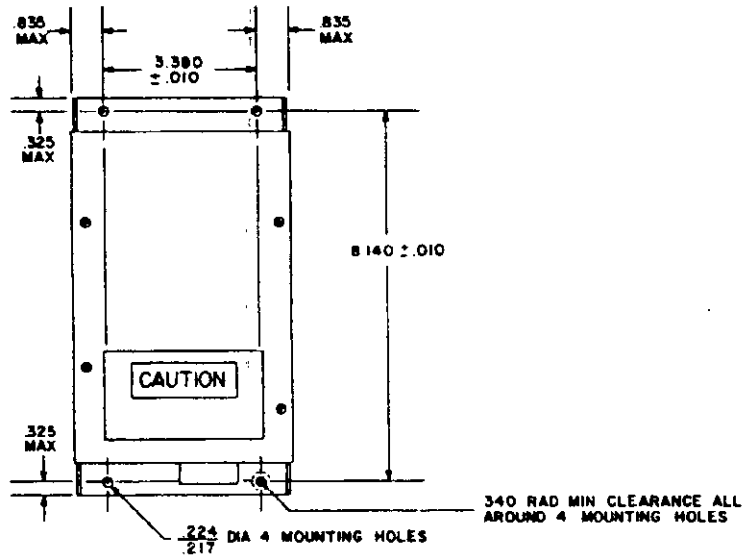


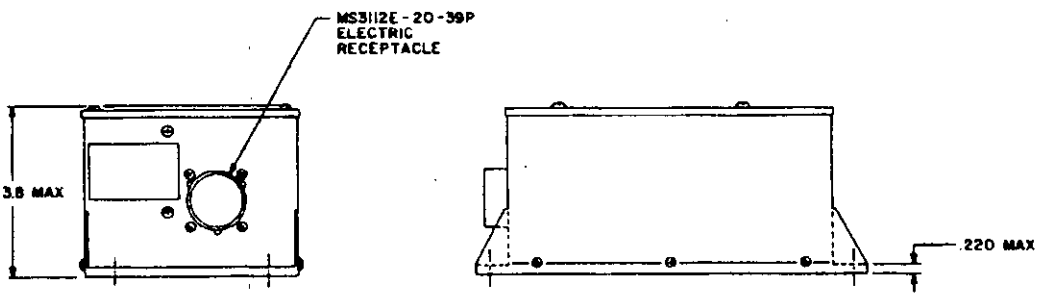


90 KVA GENERATOR

P/N M21480/19-4

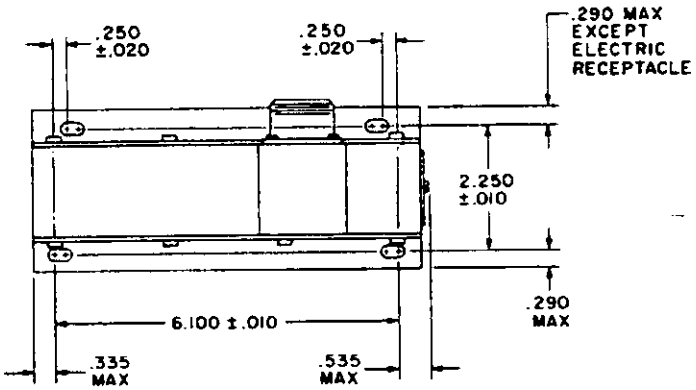
Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

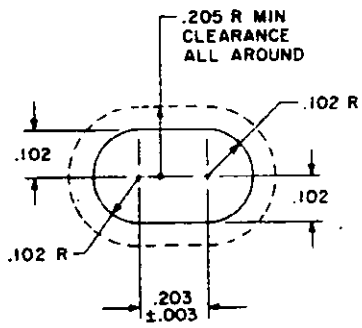




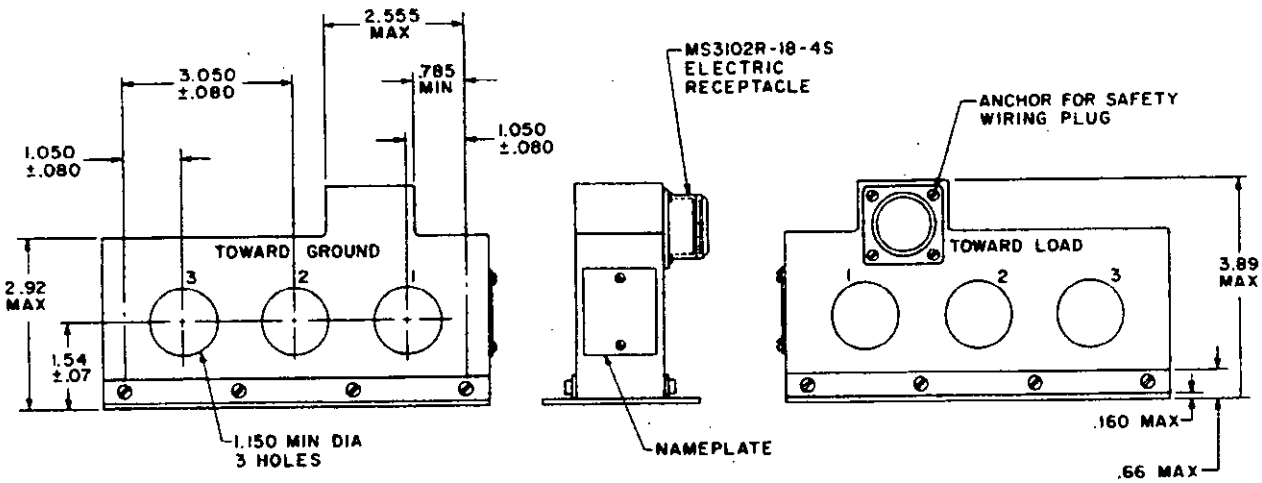
CONTROL UNIT
MAXIMUM WEIGHT 5.25 POUNDS
P/N M21480/19-6

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:43Z
Check the source to verify that this is the current version before use.





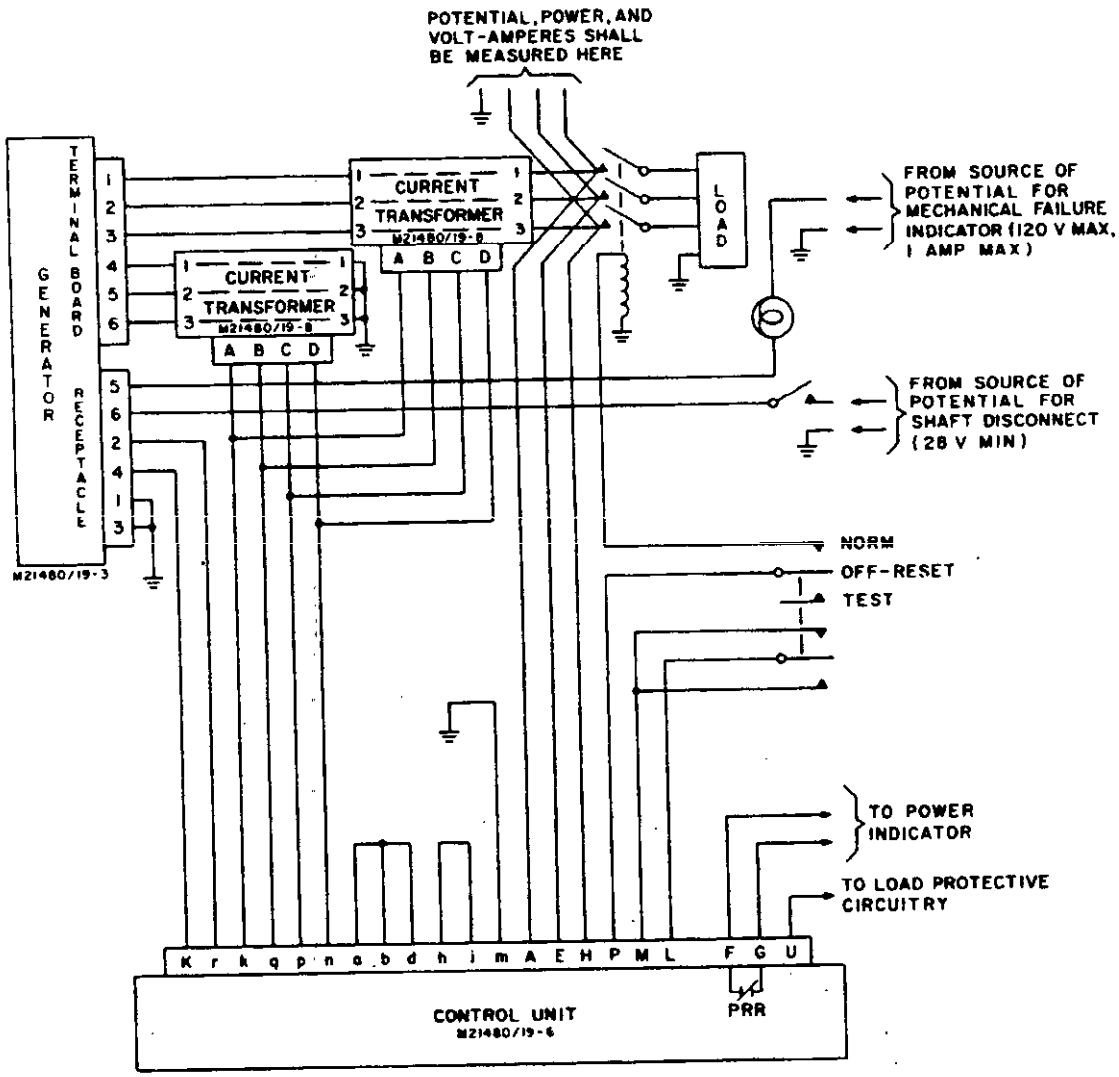
DETAIL OF
MOUNTING HOLE



CURRENT TRANSFORMER

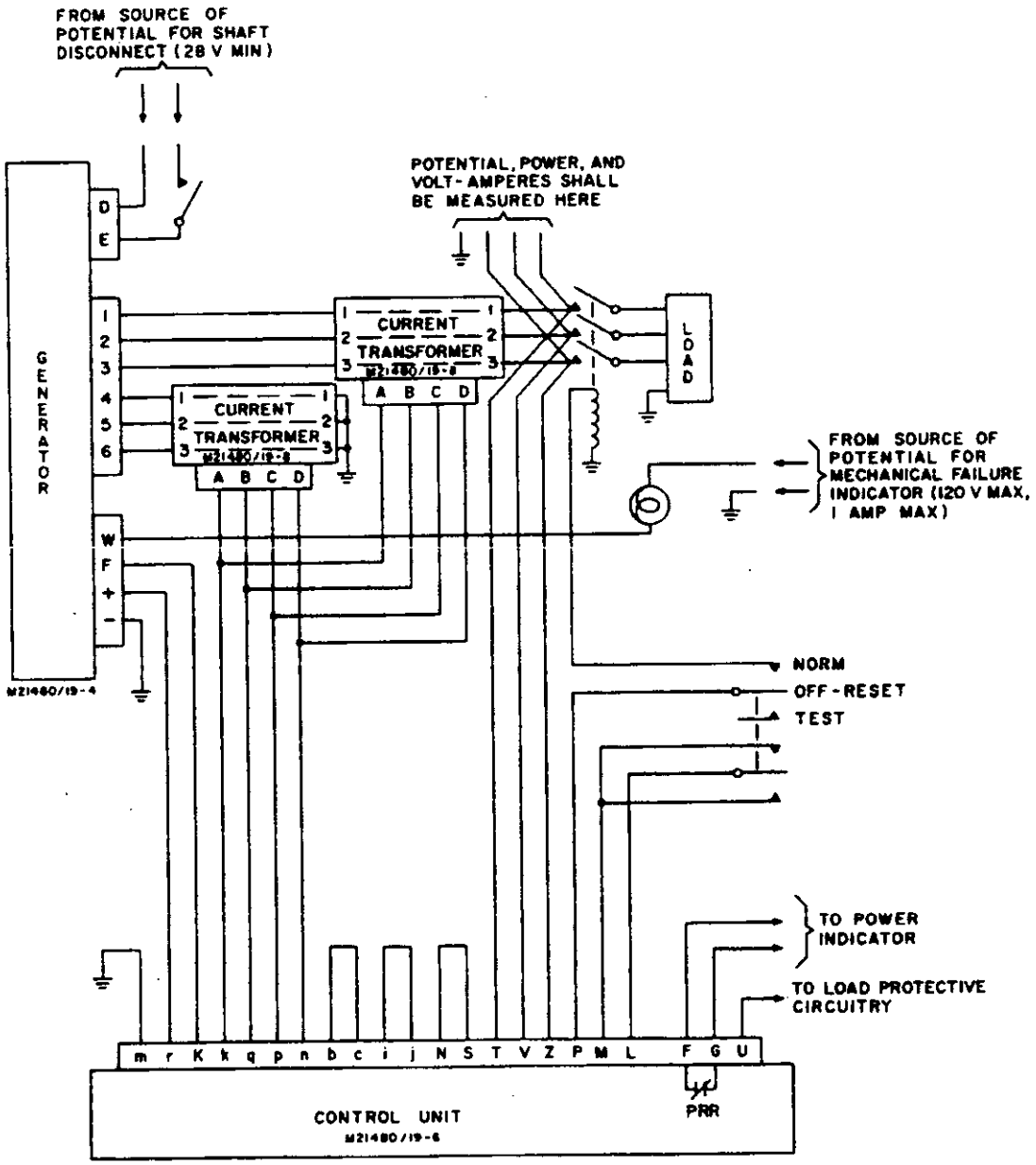
MAXIMUM WEIGHT 2.65 POUNDS
P/N M21480/19-8

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.



TEST CIRCUIT FOR 50 KVA SYSTEM

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.



TEST CIRCUIT FOR 90 KVA SYSTEM

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

1. Specification. The systems and parts shall conform to MIL-G-21480 modified as follows.

3.4.2 through 3.4.2.8 Environmental Requirements - replace with:

3.4.2 Altitude and temperature. The systems shall meet the performance requirements of this specification when carrying any load from zero to the loads shown on Figure 1 of the specification sheet in all combinations of altitude and temperature between zero altitude and the limits shown on Figure 1 of the specification sheet.

3.4.8 System capacity (rating). Replace the last two sentences with: Overload and short-circuit capacities shall be based on a rating of 40 KVA for the 50 KVA system and 60 KVA for the 90 KVA system. The effective value of the potential of each phase of each system shall conform to Figure 2 Parts 1 and 2 of the specification sheet.

3.4.8.1 Continuous capacity. Add: The steady-state voltage with balanced loads shall be 112 to 118 volts L-N for any load up to 50 KVA for the 50 KVA system and 190 to 205 volts L-L for any load up to 90 KVA on the 90-KVA system.

3.4.9.2 Flexible drive and shear section. Add: Each generator shall include means which will disconnect the generator from the driving source by shearing the shaft at the shear section or other means when energized. Each generator shall include means which shall complete a circuit to ground in the event of mechanical failure of the generator.

3.4.9.3 Electric connections. Add: The proper torque for tightening the nut on each stud shall be conspicuously and durably marked on each generator adjacent to the stud.

3.4.9.6.1 Air cooling. Change "at least four" to "at least two".

3.4.9.6.1.1 Altitude rating chart. Change "10,000 hour life" to "1,000-hour life. Change "Table II" to "Table I of the specification sheet".

3.4.10 Control component design. Replace the first sentence with: Control components shall conform to the specific requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:43Z
Check the source to verify that this is the current version before use.

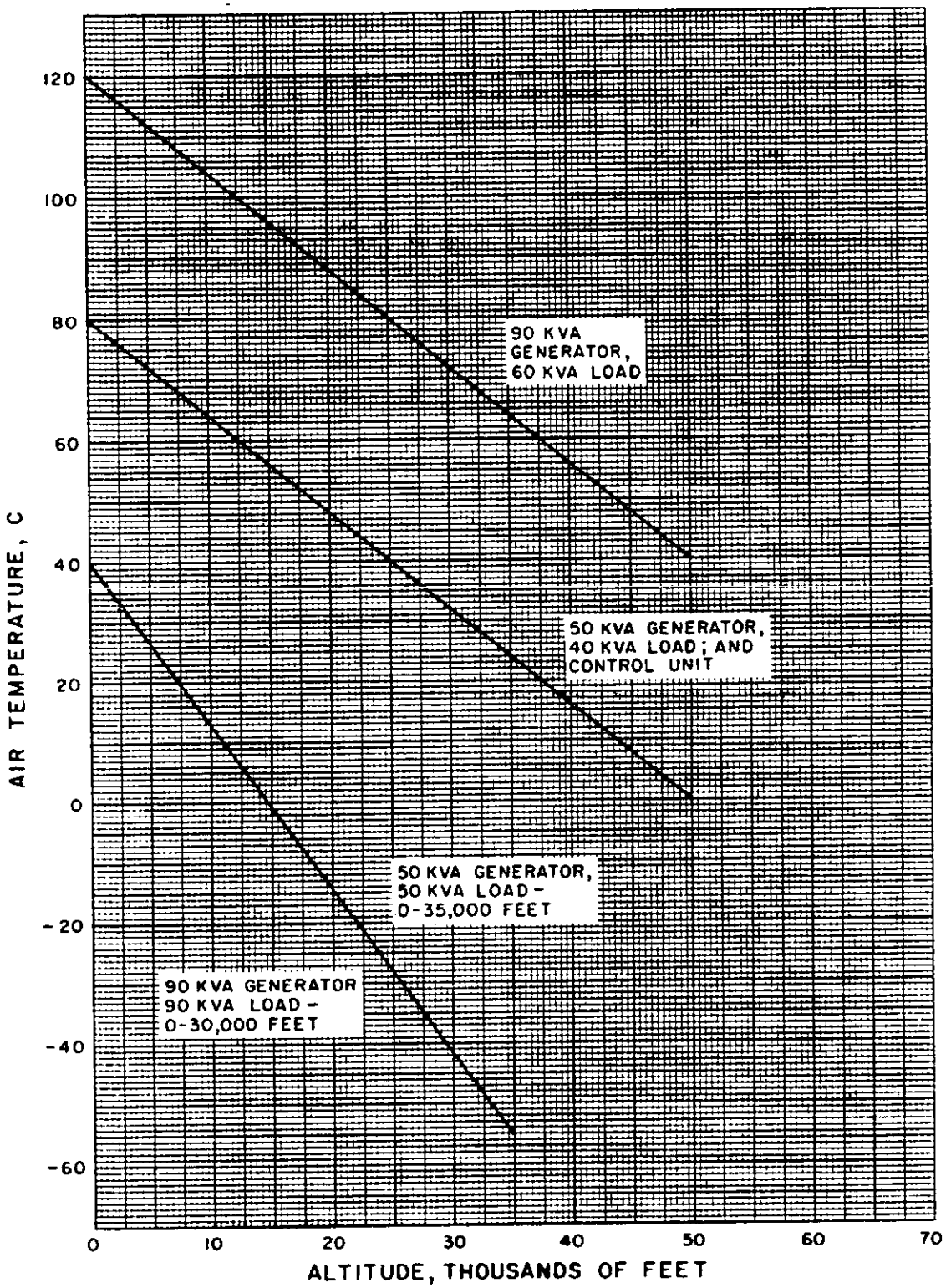


Figure 1. Temperature - Altitude Conditions for Operation.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

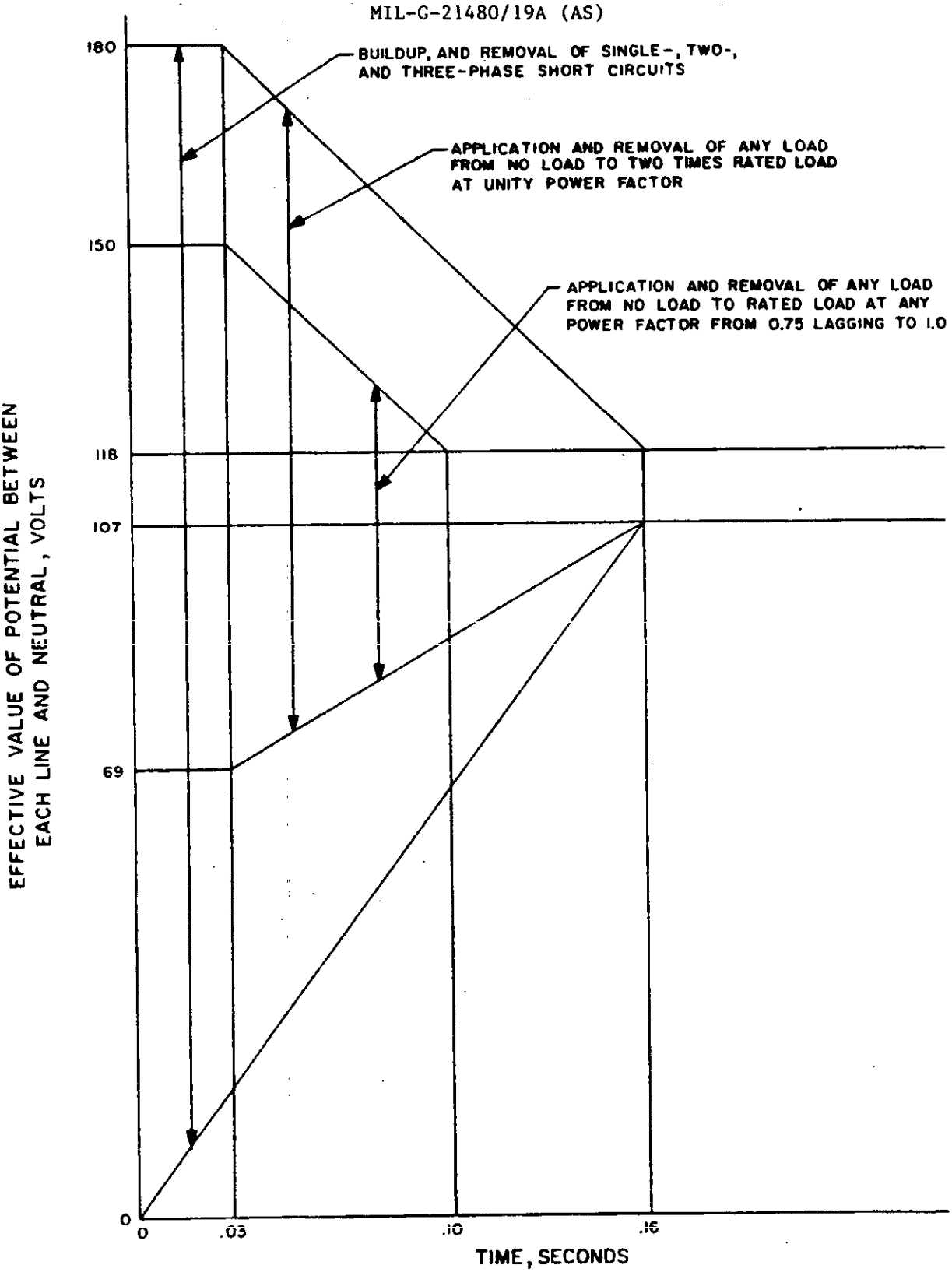


Figure 2. PART1. Transient and Steady-State Limits of Phase Potential for 50 KVA System.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

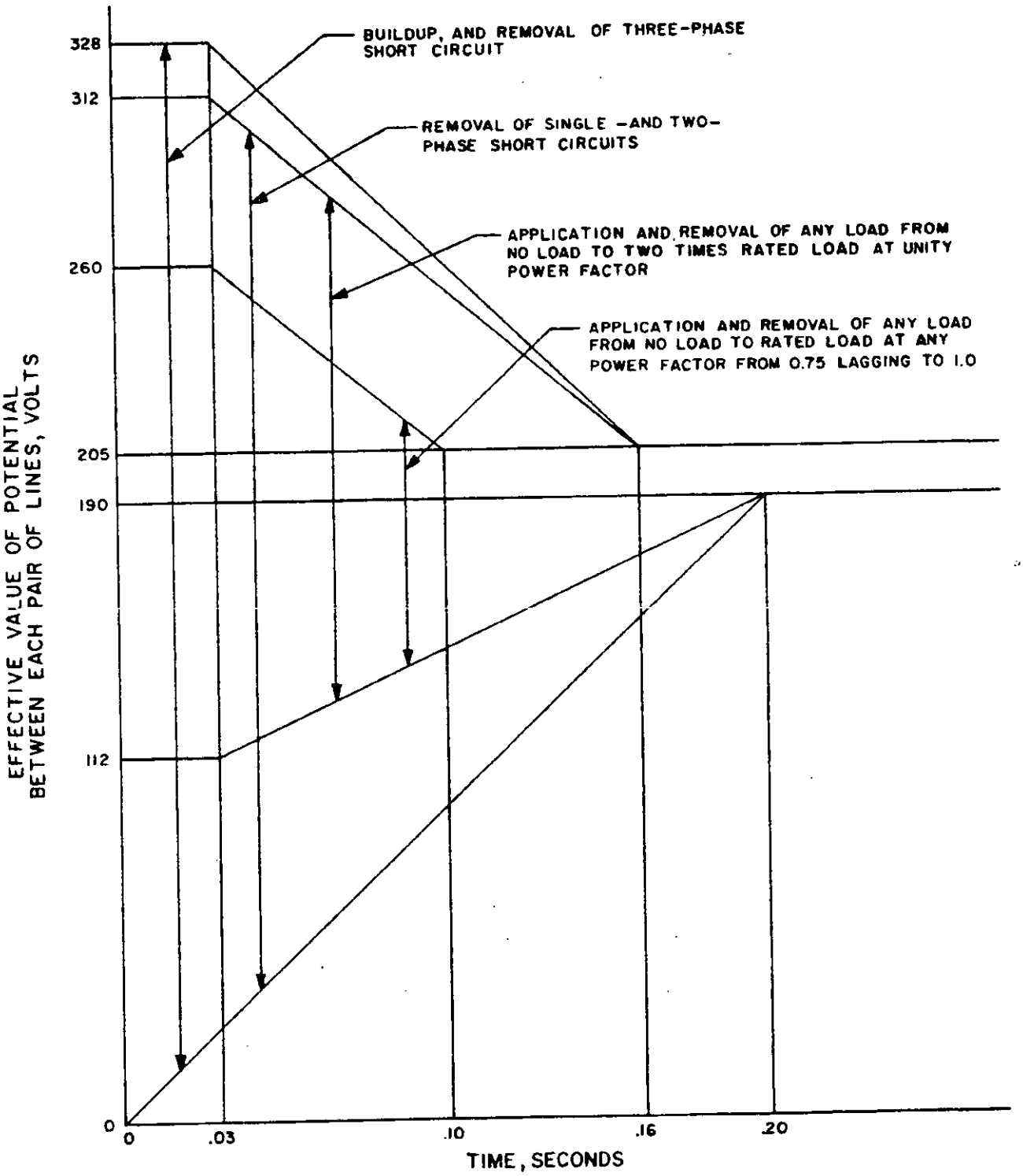


Figure 2. PART 2. Transient and Steady-State Limits of Line Potential for 90 KVA System.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

TABLE I.

Thermal Evaluation Test Points

ALTITUDE, FEET	LOAD, PERCENT OF RATED	INLET AIR TEMPERATURE, C
0	125	100 40
	100	100 80 40
35,000	125	40 0 -40
	100	40 0 -40
	50	40 0
50,000	125	0 -40
	100	40 0 -40
	50	60 0

☆U.S. GOVERNMENT PRINTING OFFICE: 1978 — 603-022/5252

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

3.4.10.6 Potting compounds. Delete.

3.4.11 Color. Change "color number 17875" to "color number 37038".

3.4.15 and 3.4.15.1 Provisions for maintainability. Delete.

3.4.16.1 Scheduled inspections. Add: The minimum number of flying hours between inspections shall be 2,000 hours. If the generator is removed for unscheduled maintenance, it shall be inspected at that time.

3.4.16.2 Scheduled maintenance. Add: Scheduled maintenance for each component shall not average more than 12 man-hours. Minimum operating time between overhauls shall be at least 2,000 hours.

3.4.17 Reliability. Replace the text with: The MTBF (Mean Time Between Failures) for each system shall be at least 2,000 operating hours.

3.5.1 Voltage limits. Change "Figure 3" to "Figure 2 of the specification sheet".

3.5.2.1 Amplitude of voltage modulation. Change "1.0 volts" to "3.25 volts".

3.5.4 Wave form. Replace the text with: The crest factor for each phase voltage waveform shall be 1.41 $\pm$ 10%. No single harmonic shall exceed 3% of the fundamental. The total harmonic content shall not exceed 5% with any load from zero to rated load.

3.5.5 Integral control power. Replace the last sentence with: This power shall be a direct potential with a value of 28 $\pm$ 4 volts with any load from 0 to 1.7 amperes.

3.5.7.2 Overvoltage. Replace the text with: Overvoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator before the voltage at the point of regulation exceeds 130 volts for 5/(V-130) seconds. Manual reset of this protective element shall be possible by putting the generator switch (see 3.4.7) in the "OFF-RESET" position (then to "ON") if the output voltage of the system is less than 130 volts. There shall be no automatic reset of this protective element.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:43Z
Check the source to verify that this is the current version before use.

3.5.7.3 Undervoltage. Replace the text with: Undervoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator within a period of 6 ± 1 seconds after any phase voltage (line-neutral) falls to 97.5 ± 7.5 volts. Manual reset of this protective element shall be possible by putting the generator switch (see 3.4.7) in the "OFF RESET" position (then to "ON") if the voltage of each phase exceeds 105 volts. Reset shall be automatic when the system is shut down. After either automatic or manual reset, the undervoltage protection element shall provide for the system to be automatically reconnected to the load bus within 2 seconds after the lowest phase voltage exceeds 105 volts.

3.5.7.4 Underfrequency. Replace the text with: Underfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator in 0.5 second after the frequency falls below 360 Hz. Manual reset shall be possible by putting the generator switch (see 3.4.7) in the "OFF RESET" position (then to "ON") if the output frequency of the system exceeds 370 Hz. After either automatic or manual reset the underfrequency protection shall provide for the system to be automatically reconnected to the bus within 2 seconds after the frequency exceeds 370 Hz.

3.5.7.5 Overfrequency. Replace "time 1 to 3 seconds" with "0.5 second". Change "525 Hz" to "435 Hz". Delete the last sentence.

3.5.7.6 Feeder fault. Change "25 amperes" to "60 amperes".

Add: 3.5.7.8 Test provisions. Each system shall include means for testing OV, UV, UF, and feeder fault protective devices, as applicable. The test controls shall consist of a selector switch and a pushbutton located on the front surface of the control panel.

3.5.8 Electromagnetic interference. Replace the text with: The systems shall meet the requirements of MIL-I-6181 except that the conducted interference test shall be performed in accordance with 4.5.12.1.

4.2.1.1 Qualification. Add: Qualification tests shall be supplemented with a minimum 100-hour failure-free flight test on an EC-130Q airplane before consideration is given to incorporating the item on the Qualified Products List.

4.4.1.1.1 Mounting generators for test condition A. Add: The M21480/19-3 generator shall be mounted on the M21480/19-5 mounting plate by means of the M21480/19-7 mounting clamp.

4.4.2.1.1 Mounting generators for test condition B. Add: The M21480/19-3 generator shall be mounted on the M21480/19-5 mounting plate by means of the M21480/19-7 mounting clamp.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.

4.4.3.3 Voltage measurement. After "line-to-neutral value", add "for the 50 KVA system, and line-to-line value for the 90 KVA system."

4.5.2 Maximum speed for regulation. Replace the last sentence with: This test shall be performed in Condition B.

4.5.3.1 System control qualification test. Change "Figure 3" to "Figure 2 of the specification sheet".

4.5.3.2 Acceptance tests. Replace "Figure 3" with "Figure 2 of the specification sheet". Replace "no load, half load, full load, and 150% load" with "no load and full load". Add: Transient tests shall be performed at 0.75 power factor for the 50 KVA system and at 0.9 power factor for the 90 KVA system. Protective functions need to be tested with only one generator.

4.5.5 Dielectric strength. Replace the last sentence of the first paragraph with: This test shall be performed in Condition B.

4.5.8.1 Phase balance qualification test. Delete all requirements for testing with 1/6 unbalanced loads. Change "0.5%" to "one percent".

4.5.8.2 Phase balance acceptance test. Delete "unity power factor". Change "0.5 percent" to "one percent".

4.5.11 Output voltage modulation. Change "1.0 volt peak to peak" to "3.25 volts peak-to-peak".

4.5.12 Electromagnetic interference. Replace "MIL-STD-461" with "MIL-I-6181 except that the conducted interference test shall be performed in accordance with 4.5.12.1".

Add: 4.5.12.1 Conducted interference. The conducted radio-interference measurements shall be made using the setup and procedures of MIL-I-6181. The conducted radio interference produced by the generator and control panel with the generator operating at minimum and maximum rated speeds at no load and at rated current, both at rated voltage, and measured between any 2 output terminals or any output terminal and ground, shall not exceed 5 millivolts over the range of 0.15 to 20 megahertz. (There shall be no external filtering).

4.5.13 Wave form. Replace the text with: The crest factor and harmonic content, line-line and line-neutral, of the output voltage shall be determined under the following conditions:

a. At generator speed of 6000 RPM for no load, and in 20% load increments to full load, unity power factor.

Source: https://assist.dla.mil -- Downloaded: 202640-19T09:43Z
Check the source to verify that this is the currentversion before use.

b. Condition (a) shall be repeated for 0.75 power factor.

c. Conditions (a) and (b) shall be repeated at generator speeds of 5700 and 6300 RPM.

The following requirements must be met during all above tests. The line-neutral crest factor shall be 1.41 ± 10%. No single harmonic shall exceed 3% of the fundamental. The total harmonic content shall not exceed 5%.

4.5.20 through 4.5.20.2 Salt Spray. Delete.

4.5.21 Fungus resistance. Delete.

4.5.22 Sand and dust. Delete.

4.5.23.1 Qualification vibration test. Replace the text with: This test shall be performed on Condition B.

4.5.23.1.2 Control components qualification vibration test. Delete.

4.5.23.2 Acceptance vibration test. Replace the text with: No acceptance vibration test is required.

4.5.24 through 4.5.24.3.4 Shock. Delete.

4.5.25 through 4.5.25.2.1 Humidity. Delete.

4.5.29 Altitude rating chart. Change "Table II" to "Table I of the specification sheet".

4.5.30 Acceleration. Delete.

4.5.32 VAST compatibility demonstration. Delete.

4.5.33 through 4.5.33.2 Reliability and maintainability testing. Delete.

NOTES:

1. Dimensions. All linear dimensions are in inches except roughnesses, which are in microinches. All tolerances unless otherwise specified are: linear, ± .030; angular, ± 1°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:43Z
Check the source to verify that this is the current version before use.

IDENTIFICATION OF ITEMS	
ITEM	MILITARY PART NUMBER
50 KVA SYSTEM (CONSISTS OF: ONE M21480/19-3 GENERATOR ONE M21480/19-5 MOUNTING PLATE ONE M21480/19-7 MOUNTING CLAMP ONE M21480/19-6 CONTROL UNIT TWO M21480/19-8 CURRENT TRANSFORMERS)	M21480/19-1
90 KVA SYSTEM (CONSISTS OF: ONE M21480/19-4 GENERATOR ONE M21480/19-6 CONTROL UNIT TWO M21480/19-8 CURRENT TRANSFORMERS)	M21480/19-2
50 KVA GENERATOR	M21480/19-3
MOUNTING PLATE FOR 50 KVA GENERATOR	M21480/19-5
MOUNTING CLAMP FOR 50 KVA GENERATOR	M21480/19-7
90 KVA GENERATOR	M21480/19-4
CONTROL UNIT	M21480/19-6
CURRENT TRANSFORMER	M21480/19-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:43Z
Check the source to verify that this is the current version before use.

GENERATOR DATA		
	50 KVA GENERATOR	90 KVA GENERATOR
RATED VOLTAGE	115-200	115-200
RATED OUTPUT	50 KVA	90 KVA
PHASE	3	3
SPEED RANGE	5700-6300 RPM	5700-6300 RPM
MAX SPEED FOR REGULATION	6300 RPM	6300 RPM
OVERSPEED	9500 RPM	9500 RPM
RATED POWER FACTOR	.75 LAGGING TO 1.0	.75 LAGGING TO 1.0
EFFICIENCY MIN AT RATED LOAD	80%	80%
MAX WEIGHT	70 LBS.	125 LBS.
OVERHUNG MOMENT, MAX	450 IN-LBS	850 IN-LBS
FREQUENCY	380/420 HZ	380/420 HZ
SHEAR	4400 IN-LBS MAX	6500 IN-LBS MAX

Preparing Activity
Navy - AS
Project No. 6115-N445

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:43Z
Check the source to verify that this is the current version before use.